

Masatake Kadono

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PROFESSIONAL SUMMARY

Mechanical Engineer with experience in mechanical design review, CAE-based structural and thermal analysis, bearing systems, rotating machinery, engineering drawings, and technical documentation. Experienced in reliability-focused mechanical systems for safety-critical applications, including bearing design studies for nuclear power plant equipment and compact rotating machinery development. Interested in applying CAD modeling, simulation, troubleshooting support, and documentation skills to ride and show equipment reliability, inspection, and enhancement work.

EDUCATION

Waseda University - Tokyo, Japan

Master of Engineering in Mechanical and Aerospace Engineering

Relevant Coursework: Mechanical Design, Computer-Aided Engineering, Finite Element Analysis, Fluid Mechanics, Mechanical Vibrations, Bearing Engineering

EXPERIENCE

Mitsubishi Heavy Industries, Ltd. - Nagasaki, Japan

April 2025 - Present

Mechanical Engineer

- Conduct design studies and design reviews for bearing systems used in nuclear power plant applications, considering seismic loading, structural integrity, reliability, and long-term operational safety.
- Perform CAE-based thermal and structural simulations, including FEM modeling, mesh generation, and boundary condition definition, to evaluate stress, deformation, and temperature distribution.
- Support reliability-focused mechanical design decisions by evaluating bearing performance, heat generation, vibration behavior, and mechanical feasibility.
- Develop compact mechanical systems and rotating machinery concepts, considering shaft layout, bearing support, component arrangement, structural constraints, and manufacturability.
- Create and revise 3D mechanical models and engineering drawings using Creo Parametric for bearing-related components, rotating machinery, and mechanical assemblies.
- Prepare technical presentation materials to communicate analysis conditions, simulation results, design considerations, and engineering rationale for design discussions.
- Coordinate with engineers and related stakeholders to review technical issues, clarify design requirements, and support improvement activities.

ENGINEERING PROJECTS

Ultra-Compact Expansion Turbine Development

October 2025 - Present

- Support the development of an ultra-compact expansion turbine by evaluating rotating machinery design considerations.
- Review shaft layout, bearing support, component arrangement, structural constraints, manufacturability, and assembly feasibility for a compact rotating system.
- Create and revise engineering drawings and 3D mechanical models using Creo Parametric; evaluate mechanical feasibility from rotor support, packaging, assembly, operation, and maintenance perspectives.

Metal Mesh Damper Research

November 2025 - Present

- Conduct experimental and analytical research on metal mesh dampers as vibration-damping materials for mechanical systems.
- Evaluate damping characteristics, structural behavior, and vibration attenuation performance under different mechanical conditions.
- Analyze experimental and simulation results to assess applicability to rotating machinery and vibration-sensitive equipment; study how material structure, compression conditions, and mechanical constraints influence damping performance.

RELEVANT SKILLS

Ride / Show Engineering Relevance: Mechanical design review, reliability analysis, troubleshooting support, vibration analysis, rotating machinery, bearing systems, compact mechanical systems, safety-critical mechanical systems

CAD / Documentation: Creo Parametric, SOLIDWORKS, 3D mechanical modeling, engineering drawings, design documentation, technical presentation materials

CAE / Analysis: Creo Simulate, FEM, thermal analysis, structural analysis, vibration analysis, bearing analysis, mechanical feasibility evaluation

Programming / Data: MATLAB, Python, Excel, Power Query, numerical analysis, data processing

Engineering Knowledge: Machine elements, fluid mechanics, tribology, thermal-fluid analysis, seismic design considerations, manufacturability, component layout

Language: Japanese, native; English, intermediate

Interests: Ride and show engineering, theme park engineering, mechanical systems, animatronics, rotating machinery, vibration control, reliability improvement